

V S C D P Prasad Tallapudi

Applied AI Engineer

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SUMMARY

Applied AI Engineer with 4+ years of experience, 2.5 of them shipping LLM applications at Bristol-Myers Squibb and Sanofi. Strong in RAG, document extraction, LLM evaluation and the Python/SQL data pipelines behind them, on AWS and Azure. Co-author of ClinTrialBench (under review at EACL 2027), an LLM benchmark suite for clinical and regulatory reasoning.

WORK EXPERIENCE

Software Engineering Expert, Digital R&D, Sanofi
Hyderabad, India

Feb 2026 - Present

- **Compass:** Built a Python and React app that scores clinical trial sites on risk and workload using statistical models on clinical and operational data. Every score links back to the data behind it, and an LLM drafts visit-prep notes for monitors. Meant to replace a vendor platform, with a savings target of ~€10M.
- **Protocol extraction:** Prototyped a pipeline that turns protocol PDFs into structured JSON (visit-by-assessment schedules) using PDFPlumber, Markdown conversion and Claude. Set up a Gemini LLM-as-judge evaluation on ~30 protocols; field-level extraction accuracy is ~85%.
- **Interpretability:** Led a proof of concept applying sparse autoencoders (SAEs) to GPT-2 and Microsoft BioGPT activations to study how the models represent biomedical concepts.

Software Engineer II, Applied AI, Bristol-Myers Squibb
Hyderabad, India

May 2024 - Jan 2026

- **Safety narratives:** Built a GenAI platform that writes first drafts of clinical safety narratives for medical writers, using RAG over patient data, structured outputs, validation checks and citations back to source data. Drafting time went from ~2 hours to ~2 minutes per narrative. It replaced the eNarrator vendor tool, with estimated savings of ~\$450K a year.
- Built the data pipeline behind it: joined and cleaned adverse event, lab, disposition, EDC and tracking data into 3M+ patient-level rows, converted to LLM-ready Markdown in AWS S3. Covered 50+ studies with 1,000-10,000+ patients each.
- **Protocol Amendment Analyzer:** Built a Python tool that parses DOCX/XML protocol versions, finds what was added or removed, and summarizes the changes with GPT-4.1. Reached 90%+ accuracy against SME review and saved ~60 review hours a week across 25-40 protocols a month (~\$50K a quarter).
- Built Python and SQL ETL pipelines for Global Product Supply that pull 15+ datasets (deviations, compliance findings, CAPA plans, audits) from Redshift and Impala. A single deviation can involve ~1.5M records; the pipeline turns them into clean GenAI input and cut data prep per deviation to 1-2 minutes.
- Replaced legacy John Snow Labs NER models with Gemini and OpenAI-based extraction to pull ICD-10, ICD-11, procedure and drug codes from inclusion/exclusion criteria in 100+ protocols. This automated document generation for Trial Optimization and Central Monitoring (TOCM) and saves ~\$30K a year.
- Ran data ingestion and quality checks for 600+ study protocols and validated 4 Tableau dashboards used for Study Operations and Assessment Planning (SOAP) optimization.

Senior Systems Engineer, Cognizant Technology Solutions
Bangalore, India

Jul 2022 - Apr 2024

- Wrote Python and SQL scripts for client Medidata that run daily data quality checks on Rave EDC eCRF entries, flagging fields with wrong values or late entry across up to 10K records a day.

- Replaced a manual review process with these scripts, cutting daily checking time from ~2-3 hours to ~15 minutes (about 10 hours saved a week).
- Applied machine learning classification to tag records by category before they went into reports.
- Debugged and fixed failing data jobs, and presented results to stakeholders on Agile and Waterfall projects.

PUBLICATIONS

ClinTrialBench, co-author, under review at EACL 2027 (ACL Rolling Review)

- Suite of three benchmarks (SiteMonBench, SeverityBench, CapaBench) for clinical trial and pharma quality tasks. I built SeverityBench and CapaBench.
- **SeverityBench**: the model labels FDA-style inspection findings as Minor, Major or Critical and has to cite the facts behind the label. Scored on accuracy, quadratic weighted kappa and evidence quality.
- **CapaBench**: from a deviation narrative, the model lists the facts, finds the systemic root cause and writes corrective and preventive actions. Each part is graded by an LLM judge and scored with precision, recall and F1.
- Evaluated 22 models from OpenAI, Anthropic, Google and DeepSeek through LiteLLM, with and without reasoning modes. No single model led on all three benchmarks.

SKILLS

Languages: Python, SQL, Bash, Git

LLMs and GenAI: Claude, GPT, Gemini, DeepSeek, AWS Bedrock, RAG, prompt engineering, structured outputs (JSON Schema), tool/function calling, agentic workflows, MCP, LiteLLM, LangChain, LangGraph, PydanticAI

Evaluation: LLM-as-judge, human/SME evaluation, benchmark design, hallucination checks, precision/recall/F1, quadratic weighted kappa, LangSmith

NLP and ML: NER, information extraction, embeddings, semantic and hybrid search, FAISS, Sentence Transformers, Hugging Face Transformers, PyTorch, sparse autoencoders (SAEs), BioGPT

Data engineering: ETL, Pandas, NumPy, PostgreSQL, MySQL, MongoDB, Amazon Redshift, Apache Impala, Parquet, Arrow

Backend and cloud: REST APIs, React, Docker, AWS (S3, Bedrock), Microsoft Azure (Azure Functions), GitHub Actions, CI/CD, Linux

Visualization: Streamlit, Tableau, Power BI, Excel

Domain: Clinical trials, Medidata Rave EDC, eCRF data quality checks, safety narratives, protocol documents, FDA / ICH / 21 CFR, GxP, deviation management, CAPA, root cause analysis, ICD-10/11 coding

EDUCATION

M.Tech, Artificial Intelligence and Data Science - KIET, Kakinada (work-integrated program)	2022 - 2024
B.Tech, Electrical and Electronics Engineering - Aditya University, Surampalem	2018 - 2021
Diploma, Electrical and Electronics Engineering - APT, Anaparthi	2015 -2018

CERTIFICATIONS

[IBM Data Science Professional Certificate](#) - Innomatics Research Labs

[IBM Data Engineering Professional Certificate](#) - Coursera